EUROVIS ZURICH 21

RoomCanvas: A Visualization System for Spatiotemporal Temperature Data in Smart Homes

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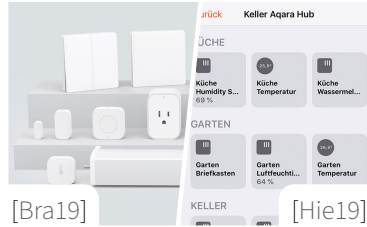
Motivation

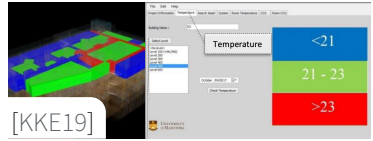
“IoT systems often involve huge amounts of data from a wide variety of sources, including sensors. […] this data must be converted into decisions and actions.” [Gar17]

- Spatiotemporal sensor data is increasingly collected and made available in smart homes
- Data is sent to and analyzed by cloud services but often not available for consumer exploration
- Unified view of sensor data and the building environment allows for variety of use cases

- Tools to explore, monitor, and understand spatiotemporal sensor data within their spatial and temporal context will become more important for experts and non-experts alike



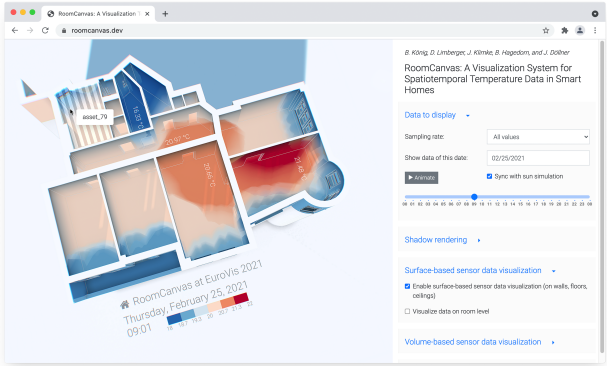
[Bra19] [Hie19]



[KKE19]

Existing systems: Often tightly coupled to expert CAD software

Demo | **Interactive, Web-based Visualization System**

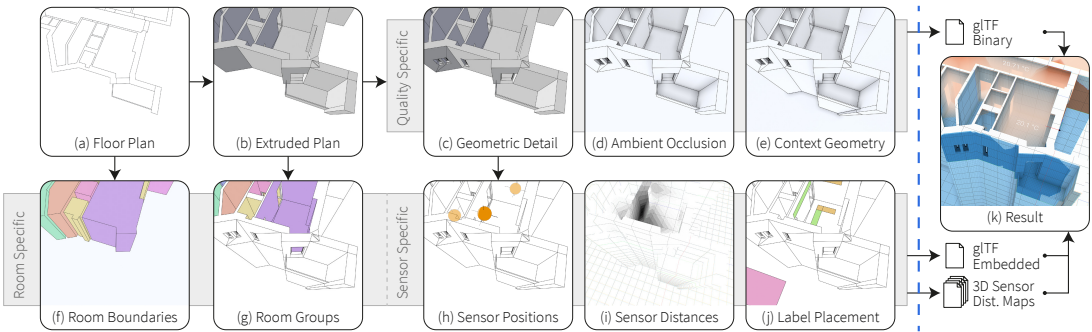


You can try it out for yourself at https://roomcanvas.dev.

Source code for both the web-based visualization and the server-side preprocessing is open sourced and available at

varg-dev / **roomcanvas**

System Details | **Overview**



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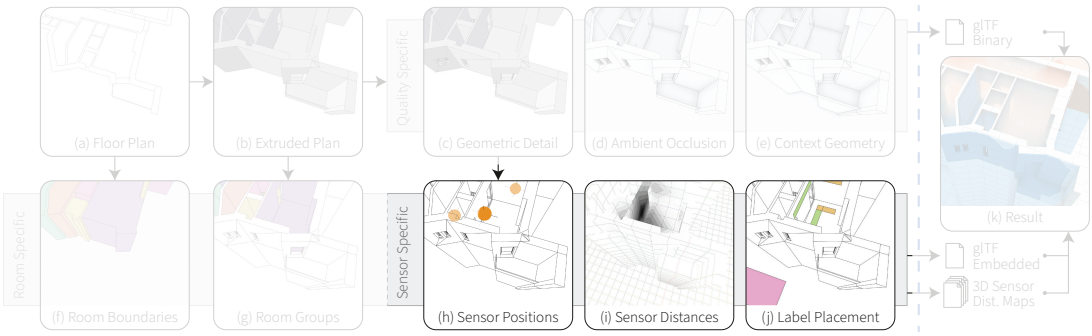
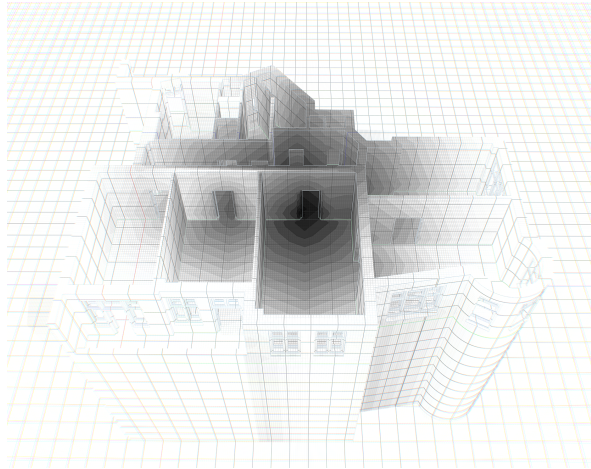


System Details | **Per-Sensor Distance Map Computation**

Precomputation of 3D distance maps for each sensor to allow for boundary-aware sensor data interpolation:



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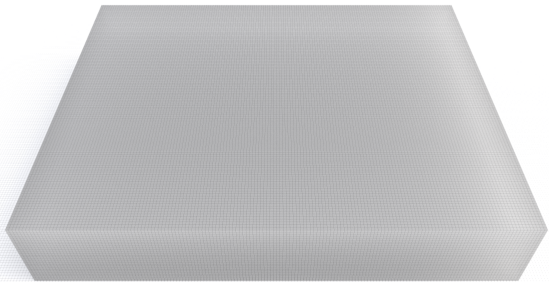

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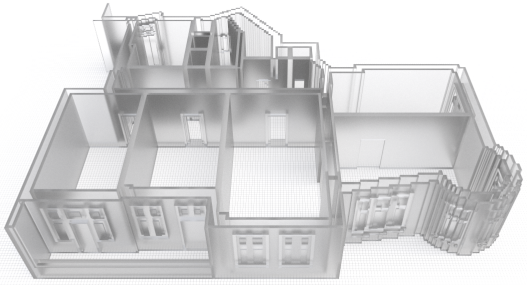

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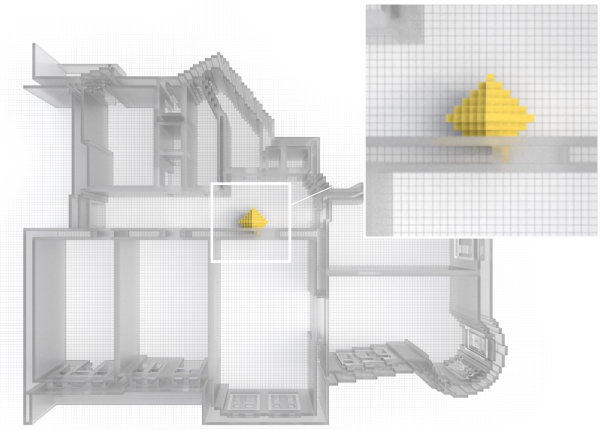
3. Use a modified flood fill algorithm to create a 16-bit distance map for each sensor, taking cells flagged as boundary into account



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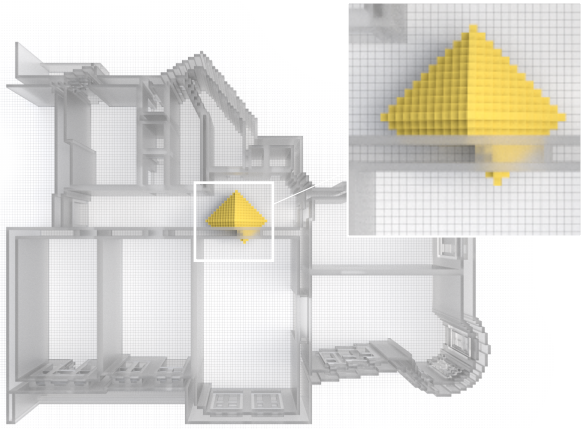

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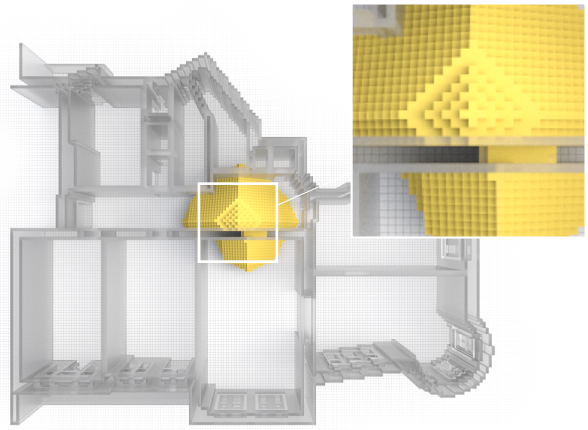
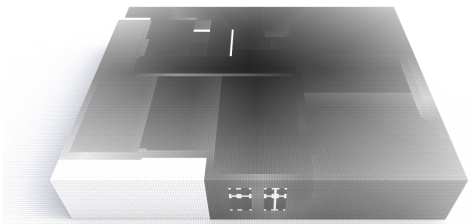
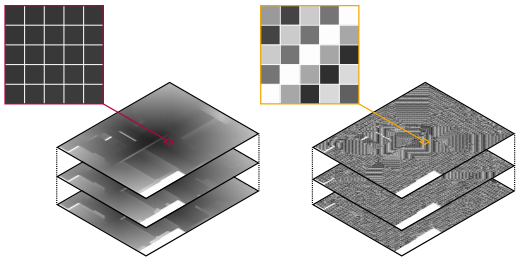
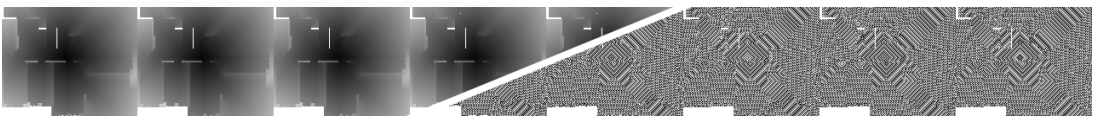


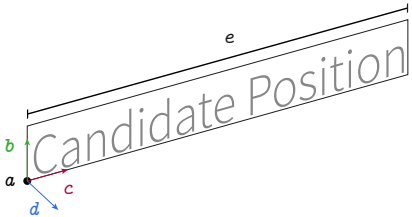
System Details | **Per-Sensor Distance Map Computation**

Due to limitations in WebAPIs regarding 3D texture loading: Encoded in two 8-bit, stacked 2D textures.



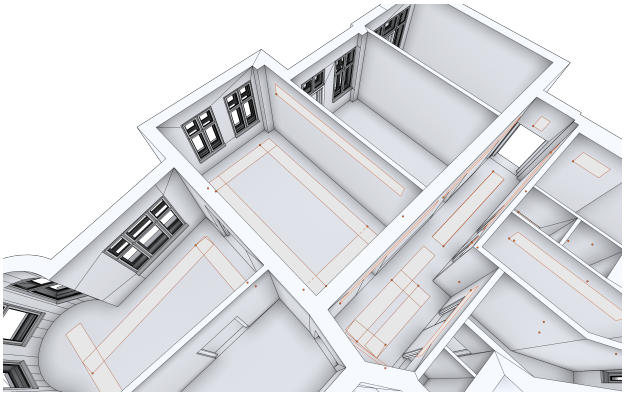






Attributes used for the algorithm:

- ***a*** candidate's anchor point,
- ***b*** upwards-facing vector,
- ***c*** running direction vector,
- ***d*** forward-facing normal vector,
- ***e*** maximum available length.

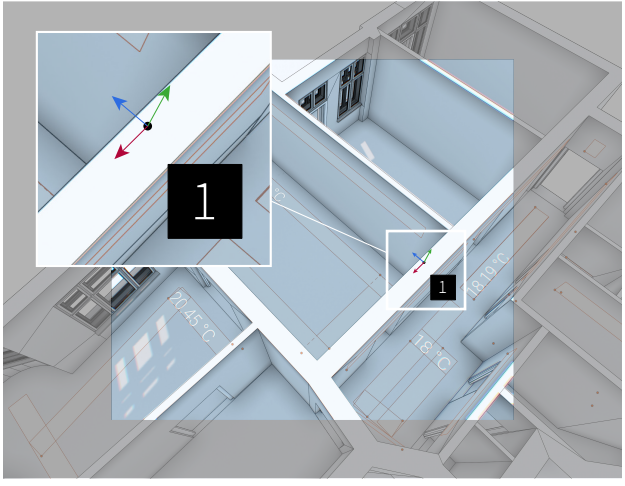


Labeling candidates are currently created manually but could be precomputed in future work.

Filtering rules:

1. Label faces camera

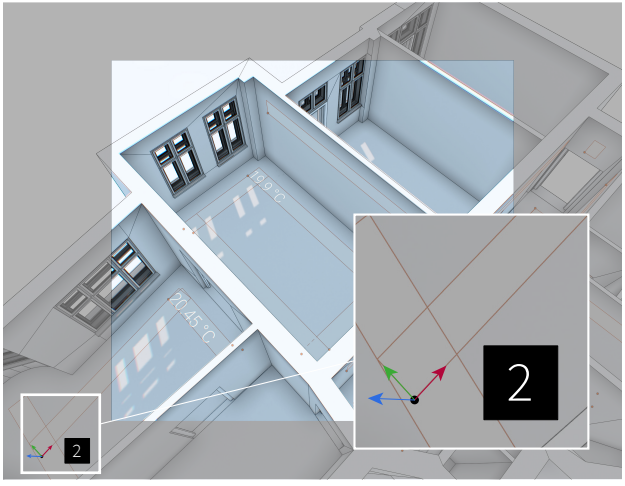
   Based on dot product between forward-facing normal vector $\boldsymbol{d}$ *and vector from camera eye to label's world coordinate* $\boldsymbol{a}$



Filtering rules:

1. Label faces camera
2. Label is inside visible viewport

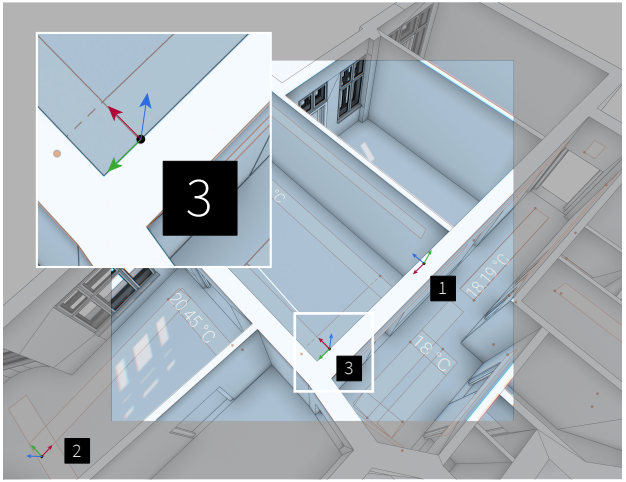
   Computed by transforming anchor point $\boldsymbol{a}$ *to normalized device coordinates (NDC)*



Filtering rules:

1. Label faces camera
2. Label is inside visible viewport
3. Label is not upside down

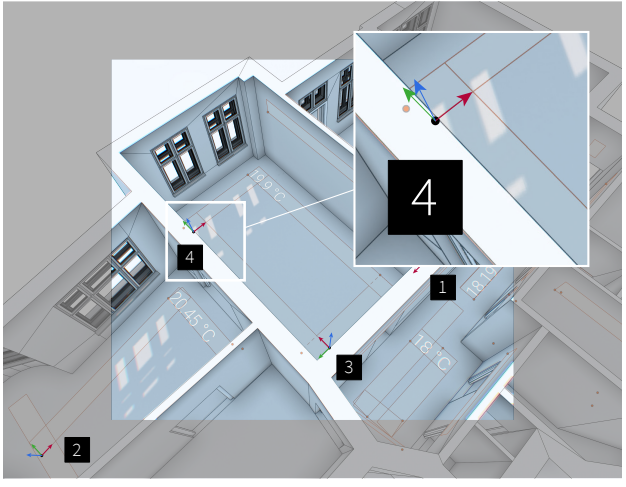
   Calculated by transforming upwards-facing vector $\mathbf{b}$ *to NDC and calculating dot product with screen's upwards-facing vector*



Filtering rules:

1. Label faces camera
2. Label is inside visible viewport
3. Label is not upside down
4. Label is not behind obstacle

Done via lookup of transformed anchor point position $\boldsymbol{a}$ (in NDC) in depth G-buffer and subsequent transformation to world coordinates



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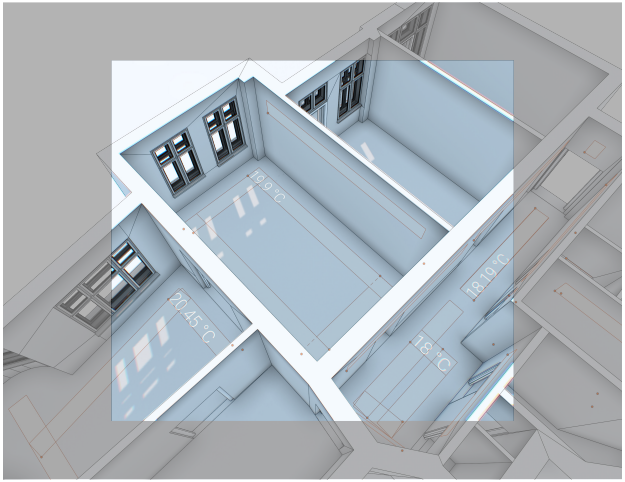
If multiple candidate positions pass filtering: Metric-driven comparison



Future Work

Potential areas for follow-up work include:

- Extension to larger buildings and industry facilities
- Optimization of sensor amount and distances [JL09]
- Simplification of the manual 3D floor plan creation

LiDAR-based capture and preprocessing of 3D floor plans

[Sin20]

Simplified 3D city model for visual embedding of the building

Questions & Comments are Welcome!

The **prototype** is available at
https://roomcanvas.dev

The **toolkit** is/will be available at
https://source.roomcanvas.dev

For further questions, comments, and collaborations feel free to contact us:

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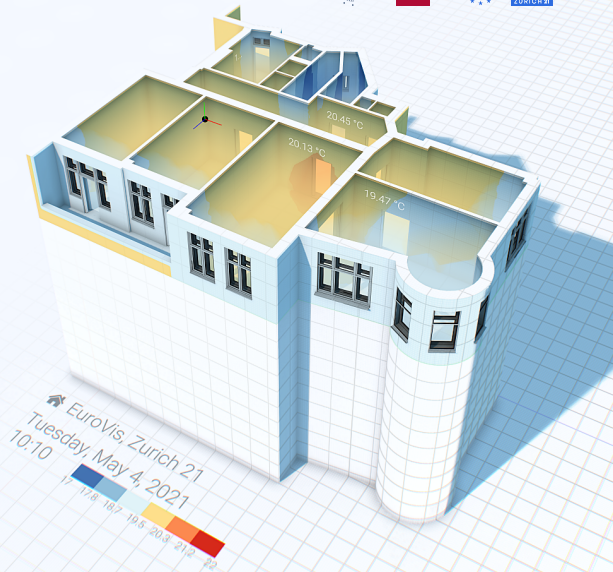

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